

Experiment 9 – Using Process Explorer to Identify Suspicious Processes

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Aim: To use Microsoft Sysinternals Process Explorer to monitor system activities and identify suspicious or potentially malicious processes running on a Windows computer.

Requirements:

- Windows Operating System
- Internet Connection
- Process Explorer (Microsoft Sysinternals Suite)
- Optional: Antivirus Software such as Windows Defender or Malwarebytes

Description:

Process Explorer is a component of the Microsoft Sysinternals Suite. It is a comprehensive tool used to display detailed information about all running processes in a Windows system. The tool assists forensic investigators and system administrators in the following ways:

- Analyzing active processes and their dependencies
- Detecting suspicious or malicious behavior in real time
- Monitoring system resource usage such as CPU and memory
- Verifying the legitimacy of processes through digital signature validation

Step-by-Step Procedure

Step 1: Download and Setup Process Explorer

1. Visit: [Microsoft Sysinternals – Process Explorer](#)
2. Click Download Process Explorer.
3. Extract the ZIP file to a folder.
4. Right-click procexp64.exe → Select Run as Administrator.

Step 2: Understanding the Interface

1. Main window shows all running processes in a hierarchical tree view.
2. Displays details like PID, CPU, Memory, and Company Name.
3. Color Codes:
 - Green: Newly started processes
 - Red: Terminated processes
 - Light Blue: Processes by current user

- Pink: Suspended processes

Process Explorer - Sysinternals: www.sysinternals.com [LAPTOP-IH45OTRE\dhyan] (Administrator)

File Options View Process Find Users Help

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
AppActions.exe		11,464 K	22,712 K	4816		
csrss.exe	< 0.01	2,300 K	4,036 K	964		
csrss.exe	< 0.01	2,944 K	4,752 K	652		
Lsalso.exe		1,236 K	2,320 K	1112		
lsass.exe	< 0.01	11,140 K	18,448 K	1132		
mc-fw-host.exe	0.17	1,16,520 K	76,568 K	5856		
mc-fw-host.exe		8,128 K	13,904 K	8520		
Memory Compression	< 0.01	1,324 K	2,72,896 K	2828		
MpDefenderCoreService.exe		11,612 K	18,652 K	5600		
MsMpEng.exe	0.17	4,07,432 K	3,34,460 K	5764		
Ngclso.exe		1,332 K	1,844 K	9876		
NisSrv.exe		4,284 K	8,012 K	13632		
Registry		13,592 K	54,476 K	184		
Secure System		176 K	43,000 K	140		
SecurityHealthService.exe		7,740 K	16,508 K	14256		
services.exe	< 0.01	5,808 K	7,808 K	1104		
smss.exe		1,164 K	712 K	632		
svchost.exe		22,412 K	50,400 K	3820		
svchost.exe		7,764 K	18,920 K	12664		
svchost.exe		4,164 K	10,744 K	14984		
System	13.54	48 K	6,136 K	4		
System Idle Process	81.23	60 K	8 K	0		
WhatsApp.exe	Susp...	35,692 K	8,864 K	17288		
WildTangentHelperService.exe	< 0.01	5,424 K	13,064 K	5668		
WindowsPackageManagerServ...		3,644 K	22,300 K	4288		
wininit.exe		1,484 K	3,280 K	656		
AppHelperCap.exe		8,496 K	11,752 K	2760		HP Inc.
BridgeCommunication.exe		5,020 K	12,740 K	1348		HP Inc.
BridgeCommunication.exe		5,996 K	14,592 K	12576		HP Inc.
CrossDeviceResume.exe		15,368 K	22,196 K	9696		Microsoft Corporation

Step 3: Identify Suspicious Processes

1. Look for unfamiliar or oddly named processes (e.g., xkdjeo.exe, abc123.exe).
2. Check Company Name and Description — genuine apps show known publishers.
3. Right-click → Properties → Image Tab → verify Path:
 - o Safe: C:\Windows\System32\
 - o Suspicious: C:\Users\<User>\AppData\Temp\ or Downloads\
4. Check Digital Signature:
 - o Valid → Trusted
 - o Invalid / None → Potential malware

Process Explorer - Sysinternals: www.sysinternals.com [LAPTOP-IH45OTRE\dhyan] (Administrator)

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Step 4: Analyze Process Behavior

1. Observe CPU, Memory, and I/O usage.
2. Unusual high resource usage → may indicate malware.
3. Right-click → Properties → TCP/IP Tab to check unknown network connections.
4. Examine Handles and DLLs Tabs for suspicious files or libraries.

Process Explorer - Sysinternals: www.sysinternals.com [LAPTOP-IH45OTRE\dhyan] (Administrator)

File Options View Process Find Users Help

Process Explorer - Sysinternals: www.sysinternals.com [LAPTOP-IH45OTRE\dhyan] (Administrator)

Image Performance Performance Graph Disk and Network GPU Graph Threads TCP/IP Security Environment Strings

Resolve addresses

Proto	Local Address	Remote Address	State
UDPv6	lapkop-in\$chte 5771	**	**
UDPv6	lapkop-in\$chte 5771	**	**
UDPv6	[fe80::0:0:27d:c029:96b5:2862]:5771	**	**
UDPv6	[fe80::0:0:5923:1267:c85b:c41]:5771	**	**
UDPv6	[fe80::0:0:3720:6646:9624:5ebc]:5771	**	**
UDPv6	[fe80::0:0:aecdd:de35:c32e:495b]:5771	**	**
UDPv6	lapkop-in\$chte 5771	**	**
UDPv6	lapkop-in\$chte 5881	**	**
UDPv6	lapkop-in\$chte 5881	**	**
UDPv6	[fe80::0:0:aecdd:de35:c32e:495b]:52240	**	**
UDPv6	[fe80::0:0:27d:c029:96b5:2862]:52241	**	**
UDPv6	[fe80::0:0:3720:6646:9624:5ebc]:52242	**	**
UDPv6	lapkop-in\$chte 52243	**	**
UDPv6	[fe80::0:0:5923:1267:c85b:c41]:52244	**	**
UDPv6	lapkop-in\$chte 52245	**	**
UDPv6	lapkop-in\$chte 52247	**	**
UDPv6	lapkop-in\$chte 52249	**	**
UDPv6	lapkop-in\$chte 56321	**	**
UDPv6	lapkop-in\$chte 56320	**	**
TCPv6	lapkop-in\$chte 5881	lapkop-in\$chte 0 LISTENING	**
TCPv6	lapkop-in\$chte 5881	lapkop-in\$chte 0 LISTENING	**
TCPv6	lapkop-in\$chte 5771	lapkop-in\$chte 0 LISTENING	**
UDP	lapkop-in\$chte 5771	**	**
UDP	lapkop-in\$chte 5881	**	**
UDP	lapkop-in\$chte 52246	**	**
UDP	lapkop-in\$chte 52248	**	**
UDP	lapkop-in\$chte 5771	**	**
UDP	lapkop-in\$chte 52239	**	**

OK Cancel

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
ETDControl.exe		6,640 K	6,240 K	6004	ETD Control Center	ELAN Microelectronics Corp.
ExpressVPNNotificationService...		33,100 K	25,640 K	16555	ExpressVPN Notifications	ExpressVPN
ExpressVPNService.exe		12,336 K	16,376 K	5472	ExpressVPN Service	ExpressVPN
chrome.exe	< 0.01	91,500 K	2,01,460 K	36708	Google Chrome	Google LLC
chrome.exe		2,208 K	6,644 K	12234	Google Chrome	Google LLC
chrome.exe	< 0.01	2,87,876 K	2,85,172 K	8104	Google Chrome	Google LLC
chrome.exe	< 0.01	20,120 K	41,636 K	18680	Google Chrome	Google LLC
chrome.exe		9,000 K	15,324 K	16176	Google Chrome	Google LLC

CPU Usage: 6.45% Commit Charge: 65.24% Processes: 240 Physical Usage: 79.80%

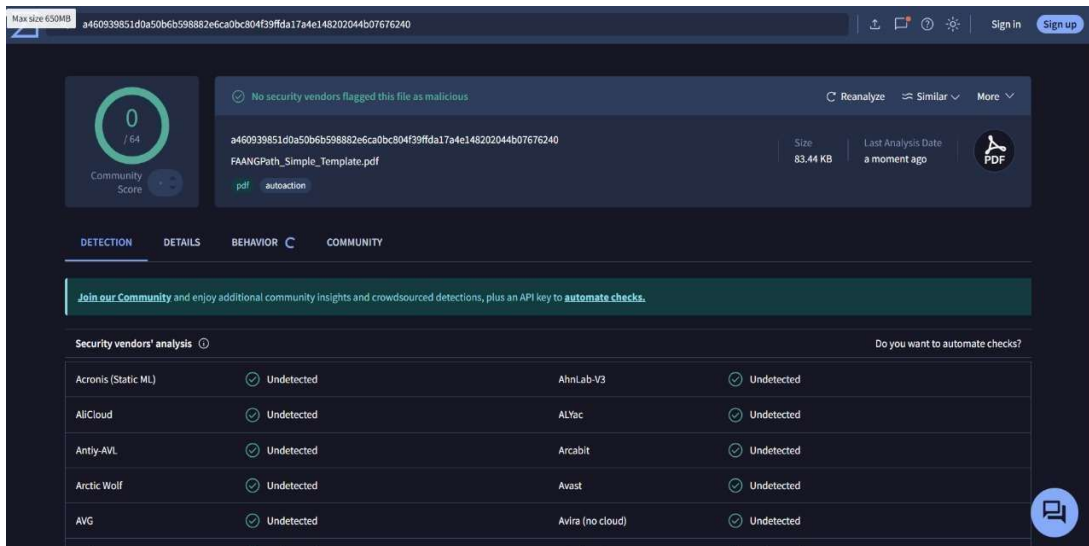
24°C Mostly cloudy

Search

ENG IN 22:42 20-10-2025

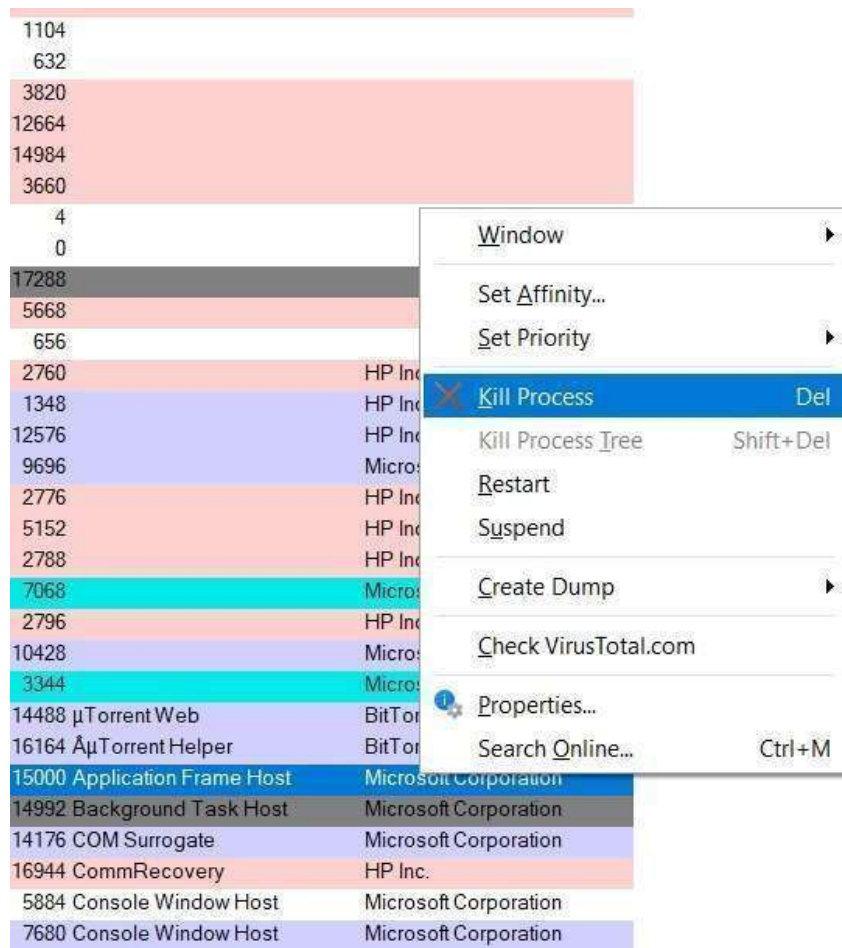
Step 5: Verify Process Legitimacy

1. Google the process name — e.g., svchost.exe vs svhost.exe (typo = malware).
2. Visit 病毒 [VirusTotal](https://www.virustotal.com) to scan or check the file.
3. Cross-check using ProcessLibrary.com or official vendor websites.



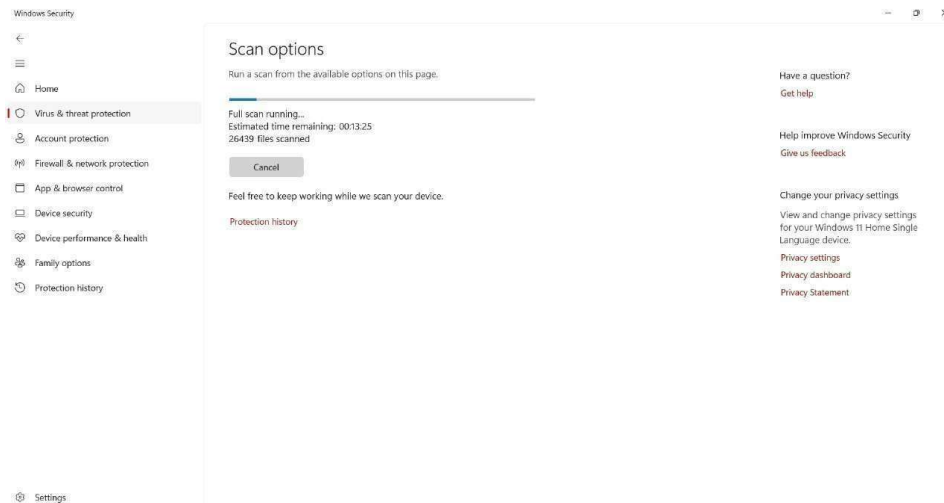
Step 6: Take Appropriate Action

- If Malicious:
 - o Right-click → Kill Process
 - o Delete the file from its directory
- If Unsure:
 - o Right-click → Suspend Process for further investigation
- After Removal:
- Perform a Full System Scan with antivirus tools



Step 7 : Example Obseravation

#	Detail	Observation
1	Process Name	faangpath_simple_template.pdf
2	CPU Usage	70%
3	Path	C:\Users\Admin\AppData\Temp\faangpath_simple_template.pdf
4	Digital Signature	None
5	Company Name	Unknown
6	Network Activity	Connected to unknown IPs
7	Online Check (VirusTotal)	Detected as Trojan
8	Action Taken	Suspended Killed Deleted Full Scan Performed



Result

Using Process Explorer, suspicious processes were identified based on their:

- CPU usage
- File path
- Digital signature
- Network activity

Malicious processes were terminated and removed, ensuring system integrity and security.

Conclusion

Process Explorer is an essential forensic tool that offers detailed visibility into active processes. It enables cybersecurity analysts and forensic investigators to:

- Detect suspicious processes
- Analyze resource behavior and signatures
- Eliminate malicious threats effectively